

AAKASH KUMAR SINGH

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Education

Indian Institute of Science, Bangalore

Master of Technology by Research in Computational and Data Sciences

CGPA: 8.5/10

Aug 2023 – Present

Maharaja Agrasen Institute of Technology, Delhi

Bachelor of Technology in Computer Science and Engineering

CPI: 8.94/10

Aug 2019 – May 2023

Publication

Aakash Kumar Singh, Dey P., Shrivatsa S., & Babu R. V. (2025). **Concept Siever: A Method for Concept Forgetting in Text-to-Image Models via Fine-tuning**. Transactions on Machine Learning Research (TMLR).

Aakash Kumar Singh, Joseph K. J., Karanam S., & Babu R. V. (2025). **Forget with Care: A Domain-Agnostic Framework for Concept Erasure without Side-effects** [Extended Abstract]. Proceedings of the U&ME Workshop at the IEEE/CVF International Conference on Computer Vision (ICCV).

Badrinath Singhal*, Aakash Kumar Singh*, Srihari K G, Ankit Dhiman, Venkatesh Babu Radhakrishnan (2025). **Sensitive Gaussians and Where to Find Them: Test Time Adaptation of 3D Gaussians**. [Under Review].

Projects

Concept Forgetting for Text to Image Diffusion Model

May 2024-August 2025

Prof. Venkatesh Babu R.

- Developed **Concept Siever**, a novel domain-agnostic framework for concept erasure in diffusion models with minimal side-effects, leading to a **TMLR publication**.
- Demonstrated **state-of-the-art** performance by improving **concept forgetting efficacy by over 33% on the I2P benchmark**; this superior result was confirmed in a user study where **our method was preferred 72.8% of the time**.
- Introduced the **first framework to offer fine-grained, inference-time concept-level control over forgetting strength**, allowing for dynamic adjustment without any retraining

Enhancing Video Diffusion with Parametric Human Models

July 2025-Present

Prof. Venkatesh Babu R.

- Researching **realistic human motion** generation within **video diffusion models** across generation, inpainting, and frame interpolation.
- Developing a novel method to enhance the realism of generated videos by **integrating the SMPL parametric human body model into the diffusion process**.
- Currently developing a **training-free method for the targeted editing of human subjects in video**, enabling precise control over their motion, location, and speed.

PEFT for Controllable Model Editing

January 2024-August 2024

Prof. Venkatesh Babu R.

- Investigated advanced PEFT techniques (e.g., AdaLoRA, SoRA) focusing on dynamic rank allocation to achieve targeted and sparse model edits.
- Developed a **system to score and identify the most influential model layers**, using these insights to design the core **targeted weight steering mechanism** for the Concept Siever framework.
- Established the technical foundation for a TMLR-accepted paper on concept forgetting, successfully bridging PEFT theory with practical applications in responsible AI.

Research Experience

Project Associate

September 2024-Present

Vision and AI Lab (VAL)

- Developed a **zero-shot motion personalisation** method for **any-to-video generative models**, enabling **realistic human motion generation** and achieving appreciable improvement in motion fidelity.
- Investigating the internal mechanisms of generative video diffusion models by applying **mechanistic interpretability** methods to map and influence the model's content generation pathways.
- Collaborated with global PhD scholars and research scientists from Adobe and Flipkart to develop novel methods for **controllable video generation**.

Achievements

Oral presentation at ICCV workshop

2025

Invited to deliver a 10-minute presentation on our extended abstract "Forget with Care" recognised by Unlearning and Model Editing in Vision Workshop at ICCV 2025.

Vision India Track

2025

Concept Siever was accepted for presentation in the Vision India Track at the Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) 2025.

Ministry of Education Post Graduate Scholarship

2023

*Secured a **national merit-based fellowship** from the **Ministry of Education** for postgraduate studies, awarded for exceptional performance in the Graduate Aptitude Test in Engineering (GATE).*

Graduate Aptitude Test in Engineering (GATE)

2023

*Secured **All India Rank 219** (Top **0.3%**) in GATE CS 2023*

Academic Service

Teaching Assistant

Indian Institute of Science, Bangalore

- DS 265o Deep Learning for Visual Analytics (August 2024 Term & August 2025 Term)
- DS 265 Deep Learning for Computer Vision (January 2024 Term)

Peer Review

2024 - Present

- Transactions on Machine Learning Research (TMLR)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Image Processing (TIP)
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

Research Mentor & Team Lead

Vision and AI Lab, Indian Institute of Science

- Guided mentees through the **full research lifecycle**, from literature review and ideation to implementation and manuscript preparation, fostering their growth into independent contributors.
- Led and mentored a cohort of 4 junior researchers (Project Associates, interns, and undergraduates), overseeing their integration into high-level Generative AI projects.